# Why impact also needs energy

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain why momentum does not tell the whole impact story.

**SUCCESS CRITERIA - I can...**

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

An impactor has 3 times the original mass at the same speed and stopping time. Compare momentum and average stopping-force size.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

**Worked reasoning:** A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

**Your turn:** Draw labeled momentum arrows and energy bars for the two carts.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Two equal-mass carts move at speeds 2 and 4 m/s. Predict how their momenta and kinetic energies compare before calculating.



Same mass; 2 m/s  Same mass; 4 m/s

**Marzano check - circle:** ① ② ③

## 04 Build the solution

Each cart has mass 2 kg. Use $K=\frac{1}{2}\times2\times\text{speed}^2$. Find both kinetic energies.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Use the idea

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 24 N net force acts right for 3 s on a 4 kg cart initially at rest. Find impulse and final velocity.

**Marzano check - circle:** 1 2 3

### 07 Exit check

For the same mass, speed triples. By what factor does kinetic energy change?

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Find two carts with equal mass but speeds in a 1: 2 ratio. Compare momentum and energy multipliers.

**Marzano check - circle:** 1 2 3